

ABC Phones Credit Portfolio Analysis

Data engineering, portfolio health, NPS, and source-data improvement recommendations

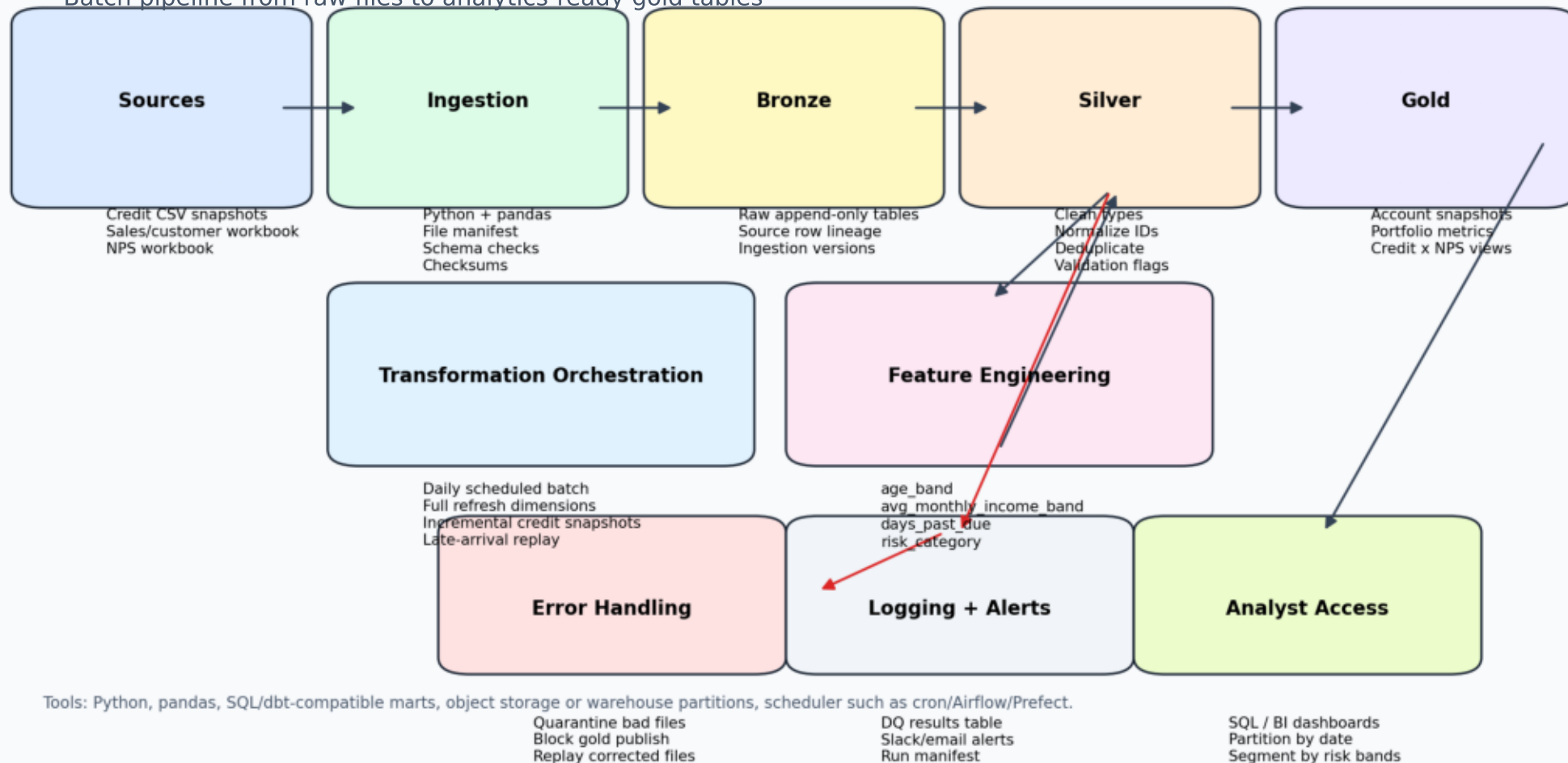
This deck summarizes a reproducible ETL and analytics workflow built from credit snapshots, sales/customer data, and NPS survey responses.

Prepared for Annex Technologies Limited Data Engineer Case Study

Data Foundation and ETL Batch ETL Architecture

Source files -> raw ingestion -> cleaning and enrichment -> quality gate -> analytics-ready gold tables

Batch pipeline from raw files to analytics-ready gold tables

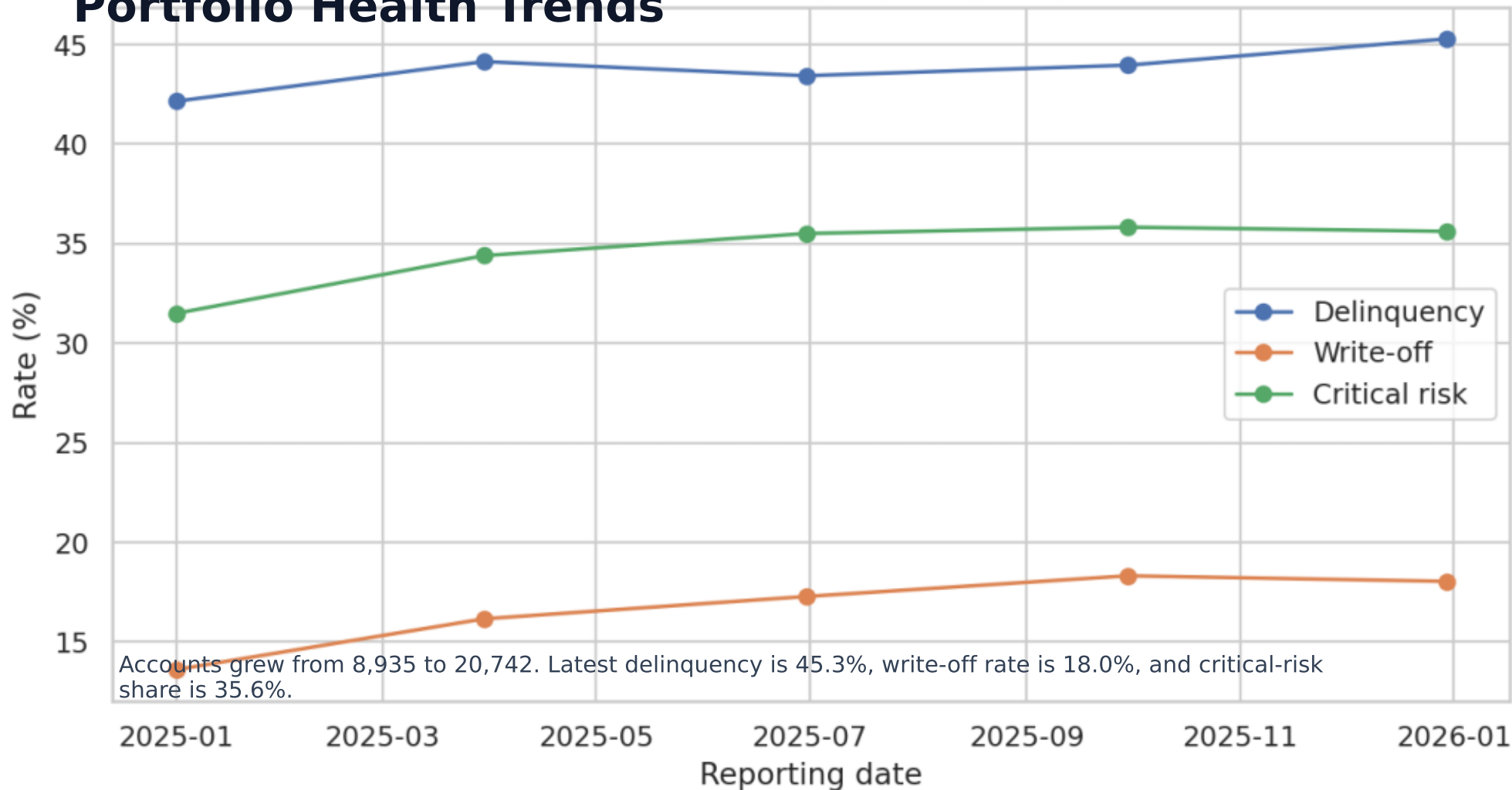


Data Quality and Feature Engineering

- `loan\_id` is the primary join key across credit, customer, and NPS sources.
- DOB and income records are sparse, so missing values are flagged rather than imputed.
- Required features created: age_band, avg_monthly_income_band, days_past_due, and risk_category.
- Risk category combines account status, arrears, DPD, and missed expected payment indicators.

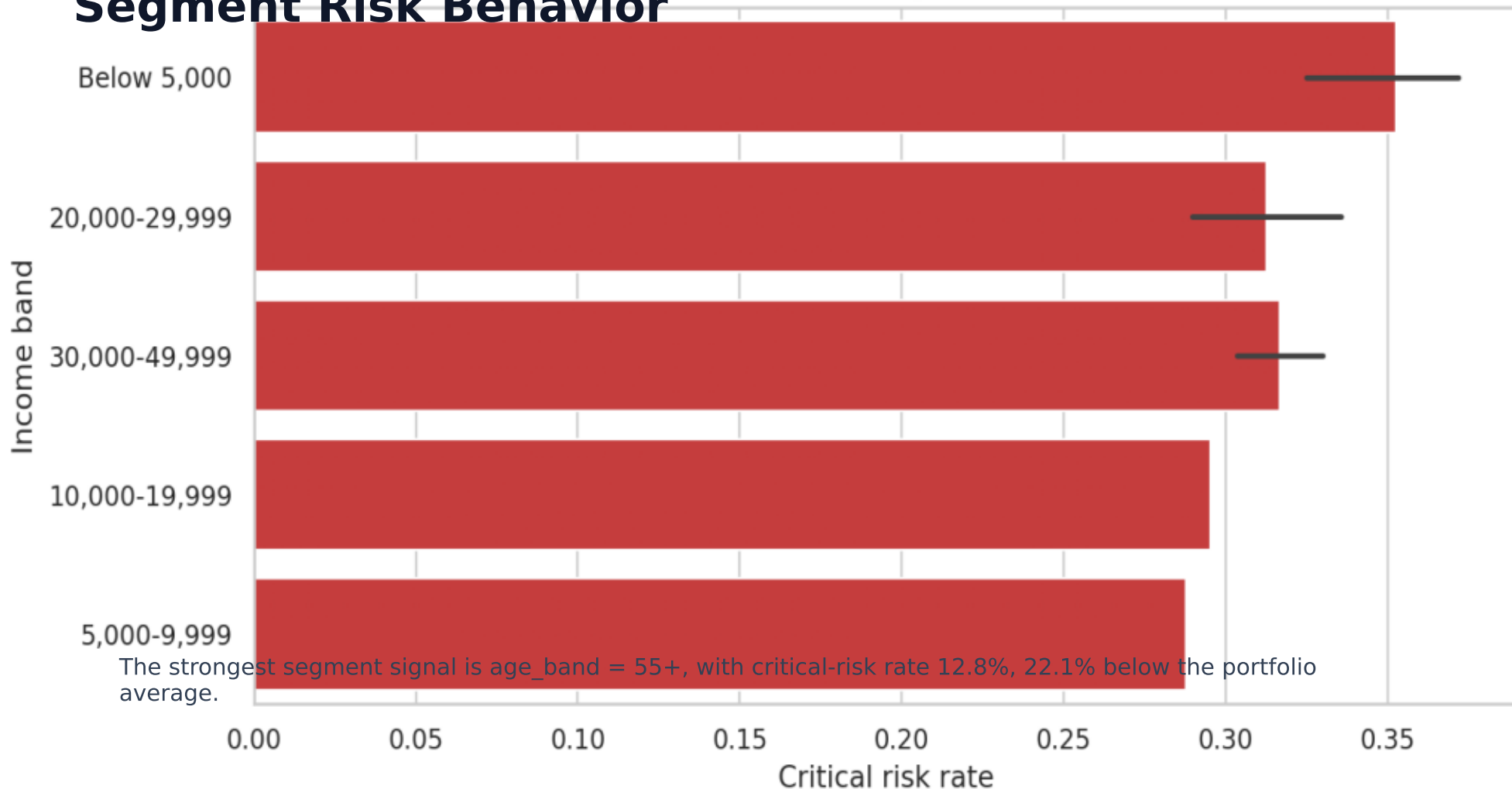
Portfolio Health Trends

Portfolio Risk Trends by Reporting Snapshot



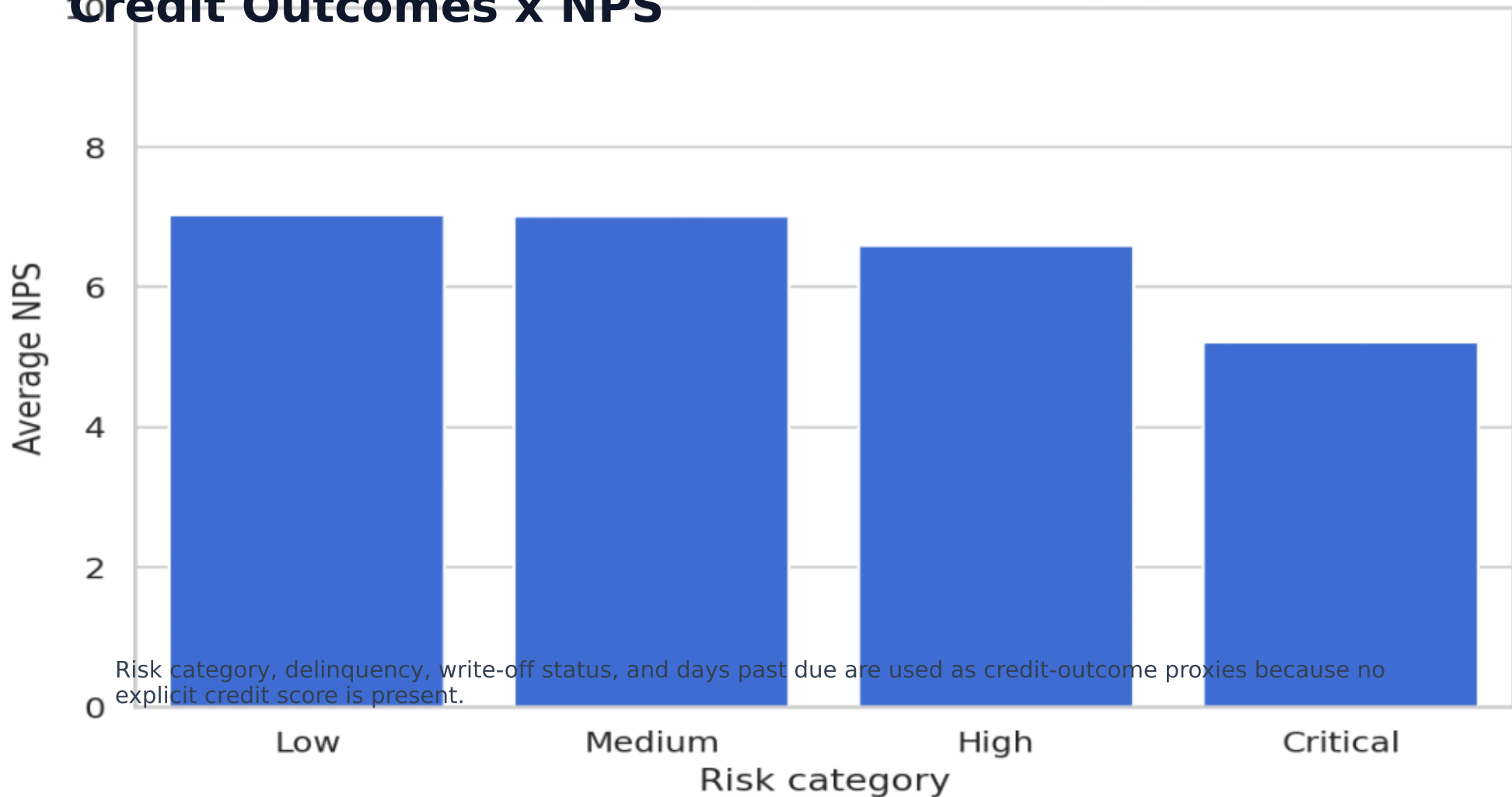
Segment Risk Behavior

Critical Risk Rate by Income Band

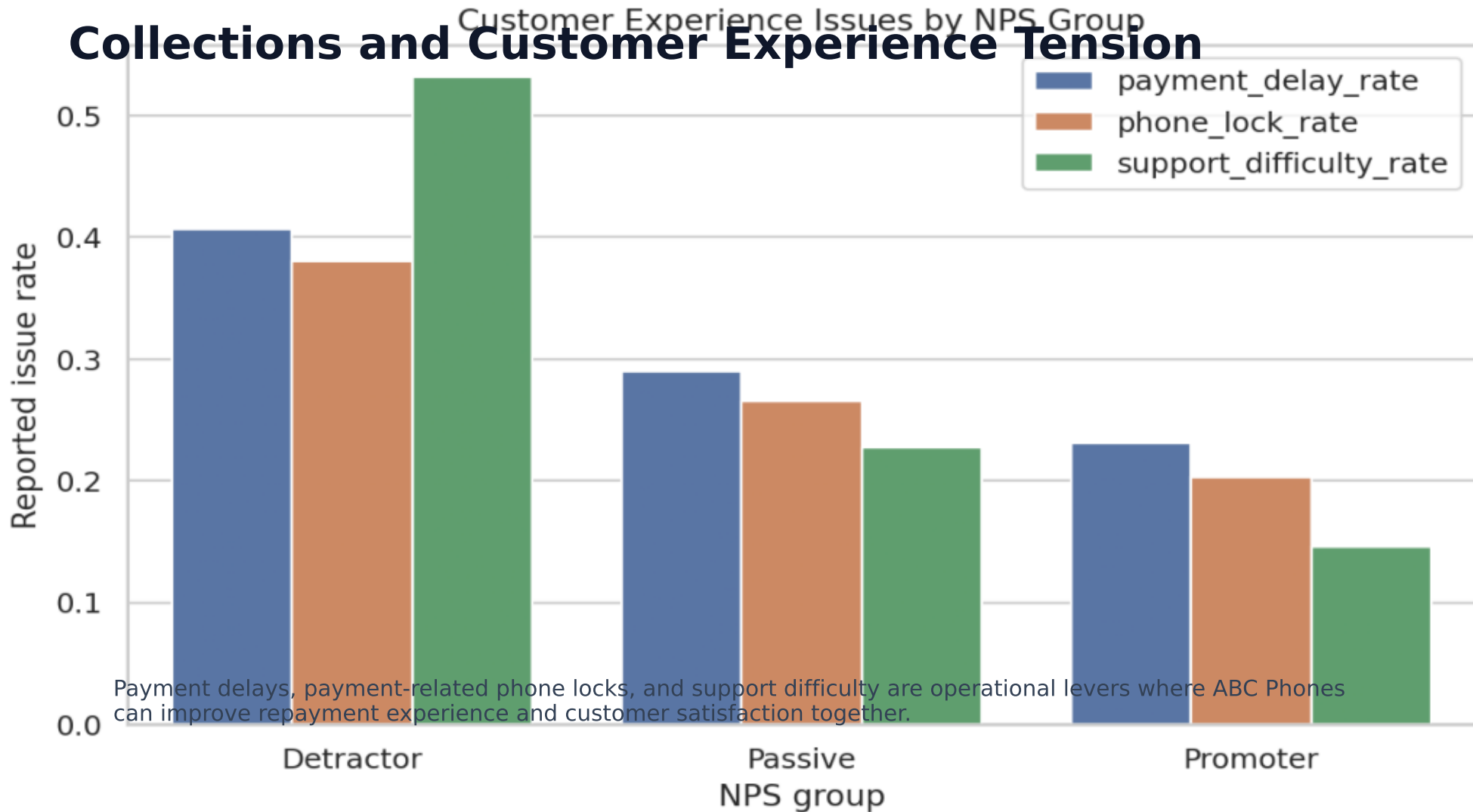


Credit Outcomes x NPS

Average NPS by Credit Risk Category



Collections and Customer Experience Tension



Data Gaps and Improvements

Missing: transaction-level payment detail, employment type, location, collection contact history, lock/unlock events, and explicit credit score.

Inconsistent: repeated customer workbook records, sparse DOB/income coverage, negative monetary values, and mixed status semantics.

Ambiguous: account status hierarchy, meaning of customer_age, and original payment due date.

Improvements: capture transaction events, govern customer/account dimensions, and publish a controlled account-status mapping table.

Recommendation

Create a payment reflection and unlock assurance workflow for delinquent or recently paid customers. Reconcile payments daily, proactively confirm reflected payments, suppress avoidable locks, and route unresolved cases to a collections-support queue.

Why this matters: it targets both repayment outcomes and the NPS pain points tied to delayed payment reflection, phone locking, and support difficulty.